Coming back to our dynamics (7.1), we notice that it can also be written as a gradient flow on $\mathbb{T}^n$:

$$\dot{\theta}(t) = n\nabla \mathsf{E}_\beta(\theta(t)),$$

for the interaction energy $\mathsf{E}_\beta : \mathbb{T}^n \to \mathbb{R}_{\geqslant 0}$ defined as

$$\mathsf{E}_\beta(\theta) = \frac{1}{2\beta n^2} \sum_{i=1}^n \sum_{j=1}^n e^{\beta \cos(\theta_i - \theta_j)}, \tag{7.4}$$

which is maximized when $\theta_i = \theta^*$ for some $\theta^* \in \mathbb{T}$ and for all $i \in [n]$. In the spirit of [LXB19], we suggest the following open problem—we recall that a critical point is called a *strict saddle point* of E_β if the Hessian of E_β at these points has at least one positive eigenvalue.

Problem 3. *With the exception of the global maxima, are all critical points of* E_β *strict saddle points?*

The proofs of Theorems 4.3 and 5.1 already yield a positive answer to Problem 5 in the regimes $\beta \leqslant 1$ or $\beta \geqslant \frac{n^2}{\pi^2}$. The complementary regime remains open. By classical arguments, recalled in Appendix A, a positive answer to Problem 3 would imply that for all initial conditions except a set of measure zero, all $\theta_i(t)$ converge under the dynamics (7.1) to a common limit as $t \to +\infty$.

Extensions of the Kuramoto model of the form

$$\dot{\theta}_i(t) = \omega_i + \frac{K}{n} \sum_{j=1}^n h(\theta_j(t) - \theta_i(t)), \tag{7.5}$$

for a general non-linearity $h : \mathbb{T} \to \mathbb{R}$, which contains both (7.2) and our model (7.1) as particular cases, have already been studied in the physics literature. For instance, we refer the reader to [Dai92] (see also [ABV+05, page 158]), where many heuristics are proposed to address the behavior of solutions to these dynamics. We are not aware of mathematical results for (7.1) besides Theorem 5.1. We nevertheless have some hope that handling the dynamics (7.1) is easier than dealing with (7.5) for a general h; for instance, we have

$$h_\beta(\theta) = e^{\beta \cos(\theta)} = \sum_{k \in \mathbb{Z}} I_k(\beta) e^{\mathrm{i}k\theta}$$

where $I_k(\beta)$ are the modified Bessel function of the first kind, whose properties have been extensively studied.

8. BBGKY hierarchy

For the sake of simplicity, we again focus on the dynamics on the circle $\mathbb{S}^1$, where recall that all particles are parametrized by angles (which we also refer to as particles). To carve out an even more complete understanding of the clustering phenomenon, it is natural to consider initial particles sampled i.i.d. from the uniform distribution on $\mathbb{S}^1$ and to study the time-evolution of the r-particle distribution $\rho_n^{(r)}(t, \theta_1, \ldots, \theta_r)$, defined as the joint law of the particles $\theta_1(t), \ldots, \theta_r(t)$. Otherwise put, it is the r-point marginal of the joint distribution $\rho^{(n)}(t, \cdot) \in \mathcal{P}(\mathbb{T}^n)$ of all n particles. Note that because of rotational invariance, $\rho^{(1)}(t, \cdot)$ is just the

uniform distribution equal to $\frac{1}{2\pi}$ for all $t \geqslant 0$. For $r = 2$, again by rotational invariance, there exists some $\psi(t,\cdot) : \mathbb{T} \to \mathbb{R}_{\geqslant 0}$ such that

$$\rho^{(2)}(t,\theta_1,\theta_2) = \frac{1}{2\pi}\psi(t,\theta_2-\theta_1).$$

Proving the clustering/synchronization of all $\theta_i(t)$ in long-time amounts to proving that $\psi(t,\cdot)$ converges to a Dirac mass centered at 0 as $t \to +\infty$. Using the fact that $\rho^{(n)}(t,\cdot)$ solves the Liouville equation, by following the method used to derive the BBGKY[12] hierarchy [GSRT13, Gol16], it is possible to show that $\psi(t,\cdot)$ satisfies

$$\begin{cases} \partial_t\psi(t,x) + \partial_x(v(t,x)\psi(t,x)) = 0 & \text{in } \mathbb{R}_{\geqslant 0}\times\mathbb{T} \\ \psi(0,x) = (2\pi)^{-1} & \text{in } \mathbb{T}, \end{cases} \tag{8.1}$$

where

$$v(t,x) = \frac{2}{\beta n}h'_\beta(x) - \frac{2(n-2)}{\beta n}g(t,x),$$

and

$$g(t,x) = \mathbb{E}\Big[-h'_\beta(\theta_3(t))\,\Big|\,\theta_1(t)=0,\ \theta_2(t)=x\Big].$$

Note that the equation (8.1) is not closed since $g(t,x)$ depends on the 3-point correlation function. This is typical in the BBGKY hierarchy, whereupon physical theory and experimental evidence is typically used to devise an ansatz for closing the system. For instance, the Boltzmann equation is derived from the BBGKY hierarchy by assuming the *molecular chaos hypothesis (Stosszahlansatz)* at the level of $r = 2$. We suggest to close (8.1) in a way that reflects the formation of clusters:

Problem 4. *Devise a realistic ansatz for $g(t,x)$ which allows to close equation (8.1), and allows to prove the convergence of $\psi(t,\cdot)$ to a Dirac mass centered at 0 as $t \to +\infty$.*

The derivation of a BBGKY hierarchy when $d > 2$, as well as for (SA), are also problems which we believe merit further investigation.

9. General matrices

Figure 5 hints at the likelihood of the clustering phenomenon being significantly more general than just the case $Q = K = V = I_d$. However, extending our proofs to more general parameter matrices does not appear to be straightforward and is an open problem. Here we discuss some particular cases (without excluding other approaches).

9.1. **The repulsive case.** As seen from Lemma 3.7, in the repulsive case $V = -I_d$ the interaction energy E_β decreases along trajectories. Recall that the unique global minimum of E_β over $\mathcal{P}(\mathbb{S}^{d-1})$ is the uniform distribution (Proposition 3.4). In contrast, we explain in this section that many different configurations of n points may yield global minima for E_β when minimized over empirical measures with n atoms.

We thus focus on minimizing E_β over the set $\mathcal{P}_n(\mathbb{S}^{d-1})$ of empirical measures, namely sums of n Dirac masses. Rewriting E_β as

$$\mathsf{E}_\beta[\mu] = \frac{e^\beta}{2\beta}\iint e^{-\frac{\beta}{2}\|x-x'\|^2}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(x'),$$

[12]Bogoliubov–Born–Green–Kirkwood–Yvon.

it turns out that minimizing E_β over $\mathcal{P}_n(\mathbb{S}^{d-1})$ is precisely the problem of finding *optimal configurations of points* on $\mathbb{S}^{d-1}$, which has direct links to the sphere packing problem [CK07, CKM+22] and coding theory [DGS91]. For $\mu \in \mathcal{P}_n(\mathbb{S}^{d-1})$, we can equivalently rewrite E_β in terms of the set of support points $\mathscr{C} \subset \mathbb{S}^{d-1}$, $\#\mathscr{C} = n$:

$$\mathsf{E}_\beta[\mu] = \mathsf{H}_\beta[\mathscr{C}] = \frac{e^\beta}{2n^2\beta} \sum_{x,x' \in \mathscr{C}} e^{-\frac{\beta}{2}\|x-x'\|^2}.$$

In [CK07], Cohn and Kumar characterize the global minima $\mathscr{C}$ of H_β. To state their result, we need the following definition.

Definition 9.1. *Let $n \geqslant 2$. A set of points $\mathscr{C} = \{x_1, \dots, x_n\} \subset \mathbb{S}^{d-1}$ is called a* spherical t-design *if*

$$\int p(x)\, \mathrm{d}\sigma_d(x) = \frac{1}{n} \sum_{i=1}^n p(x_i)$$

for all polynomials p of d variables, of total degree at most t. The set of points $\mathscr{C}$ is called a sharp configuration *if there are m distinct inner products between pairwise distinct points in $\mathscr{C}$, for some $m > 1$, and if it is a spherical $(2m-1)$-design.*

The following result is a special case of [CK07, Theorem 1.2].

Theorem 9.2 ([CK07])**.** *Let $n \geqslant 2$. Any global minimum of H_β among $\mathscr{C} \subset \mathbb{S}^{d-1}$, $\#\mathscr{C} = n$ is either a sharp configuration, or the vertices of a 600-cell*[13]*.*

The set of sharp configurations is not known for all regimes of n, d or m (the largest m such that the configuration is a spherical m-design). A list of known examples is provided in [CK07, Table 1]: it consists of vertices of full-dimensional polytopes (specifically, regular polytopes whose faces are simplices), or particular derivations of the E_8 root lattice in $\mathbb{R}^8$ and the Leech lattice in $\mathbb{R}^{24}$. We refer the reader to [CK07] and the illustrative experimental paper [BBC+09] for further detail. A complete picture of the long-time behavior of Transformers in the repulsive case remains open.

9.2. **Pure self-attention.** An alternative avenue for conducting such an analysis which has shown to be particularly fruitful consists in removing the projector $\mathbf{P}_x^\perp$, leading to

$$\dot{x}_i(t) = \frac{1}{Z_{\beta,i}(t)} \sum_{j=1}^n e^{\beta\langle Qx_i(t), Kx_j(t)\rangle} V x_j(t) \tag{9.1}$$

for all $i \in [n]$ and $t \in \mathbb{R}_{\geqslant 0}$. In fact, in [GLPR24] we analyze precisely these dynamics, and show different clustering results depending on the spectral properties of the matrix V. We briefly summarize our findings in what follows.

9.2.1. *A review of* [GLPR24]. For most choices of value matrices V, without rescaling time, most particles diverge to $\pm\infty$ and no particular pattern emerges. To make a very rough analogy, (9.1) "looks like" $\dot{x}_i(t) = Vx_i(t)$ (which amounts to having $P_{ij}(t) = \delta_{ij}$ instead of (2.5)), whose solutions are given by $x_i(t) = e^{tV}x_i(0)$. To discern the formation of clusters, we introduce the rescaling[14]

$$z_i(t) = e^{-tV} x_i(t), \tag{9.2}$$

[13] A 600-cell is a particular 4-dimensional convex polytope with $n = 120$ vertices.

[14] The rescaling (9.2) should be seen as a surrogate for layer normalization.